

## Project Design Phase-II

### Technology Stack (Architecture & Stack)

|               |                                                |
|---------------|------------------------------------------------|
| Date          | 3 October 2026                                 |
| Team ID       | SWTID-2026-4184                                |
| Project Name  | Import Data using Transform Maps (Spreadsheet) |
| Maximum Marks | 4 Marks                                        |

#### Technical Architecture:

The deliverable includes the architecture and the component/technology information based on the supplied project source.

#### Architecture: Import Data using Transform Maps

|                |                           |                                              |
|----------------|---------------------------|----------------------------------------------|
| Source         | Employee Spreadsheet      | Structured employee data in Excel            |
| Staging        | Employee Import           | Import Set staging table                     |
| Transformation | Sample Spreadsheet Import | Transform Map with Auto Map / Mapping Assist |
| Target         | Employee Test             | Custom target table with employee fields     |
| Validation     | Transform History         | Verify transformed records                   |
| Monitoring     | Reports / Dashboard       | Employee data visibility                     |

**Table-1 : Components & Technologies**

| S.No | Component                       | Description                                                                                                                                             | Technology                                                            |
|------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1.   | User Interface                  | Users prepare employee data in a spreadsheet and interact with ServiceNow forms, Import Set, Transform Map, Employee Test table, reports and dashboard. | Excel / Google Sheets; ServiceNow UI                                  |
| 2.   | Application Logic-1             | Loads the spreadsheet into a staging/import table.                                                                                                      | ServiceNow Import Sets / Load Data                                    |
| 3.   | Application Logic-2             | Maps source spreadsheet fields to the appropriate target-table fields.                                                                                  | ServiceNow Transform Maps / Mapping Assist / Auto Map                 |
| 4.   | Application Logic-3             | Transforms the staged data and updates/creates target employee records.                                                                                 | ServiceNow Transform Engine                                           |
| 5.   | Database                        | Stores the imported employee records in the custom target table.                                                                                        | ServiceNow Employee Test table (u_employee_test)                      |
| 6.   | Cloud Database                  | Cloud database service explicitly identified in the source.                                                                                             | Not specified in source                                               |
| 7.   | File Storage                    | Source employee data is provided as an Excel file; the workflow selects the file during Load Data.                                                      | Excel file / local file selection                                     |
| 8.   | External API-1                  | External API used by the application.                                                                                                                   | Not specified in source                                               |
| 9.   | External API-2                  | External API used by the application.                                                                                                                   | Not specified in source                                               |
| 10.  | Machine Learning Model          | Machine learning model used by the application.                                                                                                         | Not specified in source                                               |
| 11.  | Infrastructure (Server / Cloud) | Application is operated within the ServiceNow platform.                                                                                                 | ServiceNow platform; deployment configuration not specified in source |

**Table-2: Application Characteristics**

| S.No | Characteristics          | Description                                                                                                                        | Technology                                      |
|------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| 1.   | Open-Source Frameworks   | The source does not identify an open-source framework used for this project.                                                       | Not specified in source                         |
| 2.   | Security Implementations | The source describes Coalesce for duplicate prevention, but does not specify authentication, encryption, IAM or firewall controls. | Coalesce; other security controls not specified |
| 3.   | Scalable Architecture    | The workflow supports bulk employee-data import and repeated imports using the Import Set and Transform Map process.               | ServiceNow Import Sets / Transform Maps         |
| 4.   | Availability             | The source does not specify load balancers, distributed servers or availability architecture.                                      | Not specified in source                         |
| 5.   | Performance              | The project is intended to import bulk employee data efficiently and accurately through the Import Set and Transform Map workflow. | ServiceNow Import Sets / Transform Maps         |

**References:**

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>